Features	Plant Cell	Animal Cell
Size and Shape	Plant cells are generally rectangular in shape.	The animal cells vary in their shapes
Cell wall	Present	Absent
Plasma membrane	Present	Present
Endoplasmic reticulum	Present	Present
Nucleus	Present and lies on one side of the cell.	Present and lies in center of the cell
Mitochondria	Present and are fewer.	Present and are numerous.
Lysosomes	Present but are very rare	Present
Centrosomes	Absent	Present
Golgi apparatus	Present	Present
Cytoplasm	Present	Present
Ribosome	Present	Present
Plastids	Present with chloroplast in them.	Plastids are absent
Essential nutrients	The plant cell can synthesize amino acids, vitamins and coenzymes which are required by them	The animal cell cannot synthesize most of the amino acids, vitamins, and coenzymes which are required by them
Vacuoles	Usually large and few central vacuoles	Usually small and numerous central vacuoles
Cilia	Absent	Certain animal cells have cilia
Ribosomes	Present	Present

1.5. Acids and bases

An acid is defined as a substance that gives off protons while a base is a substance that accepts protons as per the theory of Lowry and Bronsted. Thus, an acid is a proton (H^+) donor and a base is a proton (H^+) acceptor.

The general equation that represents the dissociation of an acid is as follows:

HA	$\Longleftrightarrow$	H^+	+	A^-
Acid		Proton		Conjugate base

An acid dissociates to form proton and its conjugate base. On the other hand, the conjugate base combines with proton to form acid. The difference between an acid and its conjugate base is the presence or absence of a proton. In general, a strong acid has a weak conjugate base while a weak acid has a strong conjugate base. For instance, strong acid HCl has weak conjugate base Cl^-, weak acid HCN has a strong conjugate base CN^-.

Unit 1.indd 25 21-05-2018 15:07:38